Project Venom

The root-agent hypothesis and an experimental research programme

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ITINERANT VIRUS MODEL

1000 → 1 → 0

A hypothesis facing the possibility of being wrong

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Before we begin

This book presents the idea of an architect who has spent twenty-eight years wondering about the origin of cancer. His proposal is that different cancers might share a root agent: a virus or a similar biological entity, capable of transforming cells and leaving consequences that continue after it has passed through. That agent is called Venom.

The purpose is to explain the idea and build an observational project around it. The journey begins with a personal story, develops through questions and analogies, and reaches a sample that progressively loses its viable cells. This is where the experimental core emerges: a thousand cells, a thousand possible hiding places; one, one; zero, none.

The opening pages describe how the pieces fit together. The lighter and the fire separate the initiator from its consequence. The beaches address context. The man alone on an island turns the question towards the agent's arrival. Enigma, the cipher machine, helps us think about transformation. The twins and the sneeze direct the search towards a transient difference.

We then enter the proposed study: what to observe, what to compare, and which results would change our interpretation. The short chapters and their four boxes allow us to revisit each step. WE KNOW brings together the starting point; WE PROPOSE expresses the idea; IT PREDICTS guides expectations; HOW WE WOULD TRY TO BREAK IT sets out how to test it.

The hypothesis awaits testing, and the project does not yet provide experimental data of its own. That is the starting point for the entire book. From here, the narrative develops the proposal in its own voice, while the experimental part brings together the conditions needed to assess it.

The voice and the materials

The personal story comes from our protagonist's recollections and the reconstruction that has been preserved. The objections of the specialist researcher and the biologist friend are recorded as they appear in that journey. Their contributions helped shape the project.

The names of the participants have been replaced by their roles in the story. The analogies are explained through their essential elements. Venom retains its name as the designation of the proposed agent. The authors of scientific papers retain their attribution in the references.

Blocks 0-32 and earlier documents remain intact in a separate master backup. This narrative edition organises their content and incorporates the project's corrections. The references allow readers to consult the scientific context and the earlier work mentioned.

The experimental proposal is presented at a conceptual level. Its technical development belongs with a specialist team, under the review and institutional conditions appropriate to that work. The book offers the idea, its questions, and the initial structure from which to begin assessing it.

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Chapter 1 Seven years and two months

Before there was a project, there was a question our protagonist could not leave behind.

His father had been a heavy smoker. When he was diagnosed with lung cancer, he had not smoked for approximately seven years. Barely two months passed between diagnosis and death. Those two measures of time became linked in his memory: seven years and two months.

The question was why then. What had happened to make the disease become visible at that moment and leave so little time? Knowing the relationship between tobacco and cancer provided a context for what had happened; he kept wondering about the trigger, the event that had set that story in motion.

Our protagonist is an architect. His work on this idea consists of building a project: looking for a relationship between the pieces, sketching possibilities, recognising a problem, and asking how it might be examined. For twenty-eight years he held on to an intuition that he is now trying to explain in an orderly way.

The intuition is this: different cancers might have the same viral agent behind them, capable of acting in different tissues and leaving different consequences. Venom is the name we would eventually give to that possible root agent: a virus or a similar biological entity.

The first comparison was with influenza. A viral cause helped him imagine a diversity of manifestations without requiring a completely different origin for each one. From there, the question broadened: lung, colon, brain, pancreas... What if some of what seemed to be separate stories shared a beginning?

A decisive distinction was still missing. At first, we were trying to explain how the same virus could turn into many cancers. The problem became clearer when we changed the relationship between the two: the agent could initiate a transformation, and the tissue could express its consequence in its own way.

Cancer would be what Venom leaves behind.

That sentence began to bring order to the project. It allowed us to think of a common origin and, at the same time, different tumours. It allowed us to separate the initiator from the cells that had changed. And it opened a question that would eventually become central to everything: if the initiator could leave, where and when would we have the best chance of finding it?

The book follows that journey. It begins with a loss, moves through images and objections, and arrives at a concrete proposal for observation. Its purpose is for the reader to understand what our protagonist imagines, why he imagines it, and how he wants to turn that idea into a question others can study.

WE KNOW

Tobacco causes cancer, and quitting smoking progressively reduces the risk [R2, R3]. The seven years and the two months belong to the personal recollection from which the project arose.

WE PROPOSE

Exploring a shared origin: Venom, a virus or a similar biological entity capable of initiating different transformations depending on the tissue.

IT PREDICTS

The idea of a common root directs us to look for some shared identity or mechanism across different tumours. The experimental chapters make that expectation more specific.

HOW WE WOULD TRY TO BREAK IT

Compare what we expect to find with alternative explanations and specify which observation would distinguish the proposed intervention.

Chapter 2 The lighter is not the fire

A motorcycle, a car, and an aeroplane offer very different images. The engine comparison helped us think about a component beneath those differences: a shared principle can produce different results when the system in which it acts changes.

Applied to the project, the tissue would contribute much of the final form. A common agent could act in different cells without making its consequences identical tumours. Within the idea, differences between outcomes no longer required a different initiator for each one.

Then came a simpler image: a lighter starts a fire. The person carrying it leaves. The fire continues.

When we return to the room, we find the effects of the fire. We can study what burned, how the flames spread, and what remained standing. To find the lighter, we may need to look elsewhere or at another time.

That was the separation we were looking for. Venom would be the initiator; the transformed cell, the beginning of a process that could continue. The tumour would be a consequence retaining its own dynamics, even after the agent had left that environment.

The biological translation is a persistent transformation. The altered cell passes some of what has changed on to its descendants. That continuity allows us to imagine growth and spread without placing the initiator inside every subsequent cell. The description of cancer and its cellular spread provides the context for this distinction [R1].

Metastasis thus found a place in the narrative. What moves and multiplies may be the transformed cell, with its altered behaviour. Venom and the consequences of Venom cease to be the same thing.

The fugitive added another piece. Someone can leave the scene of an event and later blend in with those around them. They may be far away; they may

also be nearby and go unnoticed. The conversation brought together those two possibilities: movement and camouflage.

The Itinerant Virus Model retains that image of an agent capable of staying or leaving. If some tumours were places it had already left, a search limited to a late snapshot could miss the moment that interests us.

That is why the project began paying so much attention to time. We had to think of a situation in which the presumed initiator might still be present and its conditions of refuge were changing. Instead of remaining solely in front of the aftermath of the fire, we wanted to find an opportunity to observe the lighter.

WE KNOW

Tumour cells can pass alterations on to their descendants and spread to other parts of the body [R1].

WE PROPOSE

Separating the initiating agent from its consequence: Venom could stay or leave while the transformed population continues to evolve.

IT PREDICTS

The model directs the search towards specific states and moments, because it allows the initiator's presence to change during the tumour's history.

HOW WE WOULD TRY TO BREAK IT

Examine whether the sequence attributed to Venom has temporal support and whether another cause better explains the transformation observed.

Chapter 3 Eddie is still Eddie

To move forward, we had to know who was who. On one side, Venom. On the other, Eddie: the name we gave to the cell that might harbour it.

Imagine Eddie, a normal-looking cell. Within the model, Venom can associate with it and remain camouflaged. We apparently still see Eddie. Later, the cell changes its behaviour: venomised Eddie appears.

Transformed Eddie is still Eddie; it does not become Venom.

That was the purpose of giving it a name: to distinguish the occupant from its host, and the possible initiator from the transformed cell.

That distinction allows us to move through three states: apparently normal Eddie with Venom hidden inside; venomised Eddie; and Eddie's descendants retaining the alteration. Venom's presence and the persistence of its effects become two different questions.

Camouflage was the great problem. If we look at a cell and everything seems to be part of it, what would allow us to recognise what is hidden? Always searching in the most stable state could leave us facing the same image again and again.

Then came the twins.

Two identical twins are lying on a bed. Their mother knows that one has a slight cold. She looks at them, and both appear the same. Suddenly, one sneezes. For an instant, an action reveals the difference their faces had kept hidden.

The strength of the scene lies in that instant. The mother has not needed her child to change appearance permanently. A brief manifestation has been enough for her to recognise something that had remained hidden.

Within our analogy, healthy Eddie represents the first twin; Eddie with camouflaged Venom, the second. The mother represents surveillance. The sneeze represents a transient event that gives away a difference.

The experimental translation is to observe change. Monitoring over time can capture an alteration, a redistribution, or a signal that a single image misses. The question no longer requires an agent conveniently separated out and sitting still: it begins by looking for the moment when something changes.

That was the turning point. Instead of asking only how to recognise Venom while it was perfectly housed, we began asking when it might cease to be so. What situation would make that "sneeze" more likely? What would happen if the places we regarded as its refuges began to disappear?

Eddie had allowed us to separate the agent from what it transforms. The twins now gave us a way to search: pay attention to the transition. What had seemed like a domestic conversation had just brought us closer to the experiment.

WE KNOW

Monitoring over time allows us to study changes that a static observation leaves out.

WE PROPOSE

Looking for a transient difference while the conditions that, according to the model, allow Venom to remain camouflaged are changing.

IT PREDICTS

The experimental version expects an opportunity for detection during that transition, with a pattern that can be observed again.

HOW WE WOULD TRY TO BREAK IT

Check whether the signal belongs to ordinary cellular deterioration or maintains a specific relationship with the candidate we are seeking.

Chapter 4 The beaches

If we imagine a common agent, another question immediately arises: why are its consequences distributed so differently across tissues, people, and circumstances?

The beach allowed us to discuss that difference. A ship may find one coast where it is easy to land and stay, and another where the waves force it away. The ship and the beach are distinct elements in the scene. The encounter depends on both.

In the project, the tissue is that beach. Its history, characteristics, and conditions may favour Venom becoming established, modify its action, or make it harder for it to remain. This brought age, heredity, exposures, and the body's surveillance into the idea.

The story of our protagonist's father then returned. The years of tobacco use could be imagined as a persistent transformation of the terrain. The phrase that summed up that reading was: tobacco provides the firewood; Venom strikes the match. This is the distribution of roles the model proposes for thinking about a possible trigger in altered tissue. Tobacco's causal role and the persistence of risk after quitting are part of the documented context [R2, R3].

The beach also helped us imagine the passage of time. Ageing could change the conditions of reception. An inherited predisposition could mean being born with a particular terrain. Reduced surveillance could make staying easier. Different paths could lead to a similar opportunity.

Then an important correction emerged. We were turning the favourable beach into too rigid a requirement. There had to be room for children and newborns, for those who did not fit a story of decades of deterioration. And also for that strong, solitary, isolated man whom we would later take to an imagined island.

Context could facilitate the encounter without determining it completely. A difficult coast was still a coast a ship could reach. A calm coast could remain empty. In this way, the beach spoke of opportunity and persistence, and left room for other questions: when the agent arrived, how long it had been there, and what made it act.

The conversation even produced two expectations: a favourable environment might permit a prolonged stay; a less favourable one might push towards a quicker transition. That possibility reinforced the idea of observing different states and following their evolution.

The beach had linked context to the agent hypothesis. What remained was to understand why, even with very similar beaches, one story might unfold differently from another.

WE KNOW

Tobacco, age, and the characteristics of the body form part of the context in which cancer develops [R1, R2, R3].

WE PROPOSE

Tissue can facilitate or hinder Venom's action. Favourable terrain increases opportunities within the model; the possibility of action in other terrain is also retained.

IT PREDICTS

If context influences persistence and transition, observable differences should emerge when defined tissue states are compared.

HOW WE WOULD TRY TO BREAK IT

Define which conditions each beach represents and test the expected differences before adapting the image to the results.

Chapter 5 Enigma: the machine and the message

The message lay before us. We wanted to understand how it had been written.

The answer appeared in the conversation with a name: Enigma. The cipher machine allowed us to express what we were looking for. Imagine we have many encrypted messages before us: we can study them, compare them, and look for regularities. Having the machine as well would open another path: observing what goes in, how it transforms it, and what comes out.

In the Enigma analogy, Venom takes the place of the encoding machine. Venomised Eddie takes the place of the message. Finding the agent would allow us to try to reconstruct a sequence: Eddie before, Venom's action, Eddie afterwards.

The image answered an objection from the specialist researcher. If the initiator acted and then disappeared, the disease could continue, sustained by the alterations it left behind. Finding it would then have uncertain immediate therapeutic value.

Our protagonist was asking an earlier question: what would we learn about the mechanism if we could study what set it in motion? Understanding the transformation was already a purpose. It could yield new questions, new comparisons, and, later on, new possibilities for intervention.

The objection helped organise that journey. The project wanted to locate the possible root agent to study what it does, how it acts, and what remains after it has passed through. Finding it and curing a tumour would be different stages, connected by work yet to be developed.

The concept of hit-and-run appears in the literature: an initial intervention can leave effects that continue afterwards. In 1997, Shen, Zhu, and Shenk described an experimental system in rat cells with transformation and subsequent loss of detectable viral DNA [R6]. That precedent offers a concrete

reference for thinking about the separation between the initiation and maintenance of a change.

The MVI adds its own proposal of an agent that remains or moves elsewhere. This is why it matters to distinguish what stays in the cell from what Venom does. Enigma, the message, and the signature that allows them to be connected come to occupy their own places in the project.

This analogy changed the meaning of the search. It was no longer only about pointing to something inside a tumour. It was about opening a way to study the process that would have transformed it. The question "Why find it?" thus received a clear answer: to try to understand what it did.

And that answer brought us back to the practical problem. Before studying the machine, we had to find an opportunity to observe it. Before reconstructing its intervention, we had to get it to stop blending in with its refuge.

WE KNOW

There are experimental precedents for transformation associated with transient viral intervention [R6].

WE PROPOSE

Finding Venom to study the mechanism that would connect its action with the transformation of the cell.

IT PREDICTS

Attributing an initiating role requires a temporal sequence and a connection between the candidate's action and cellular changes.

HOW WE WOULD TRY TO BREAK IT

Investigate whether the candidate accompanies the process or actually participates in the transformation attributed to it.

Chapter 6 The man on the island and the pigeon

In the conversation, we used Tarzan as an initial image of a strong man removed from our everyday surroundings. What interested us about that image was isolation. Imagine that man living alone on an island. He is isolated from other people and from many of the influences we usually associate with our way of life. The scene removes him from tobacco, alcohol, everyday pollution, and routine contact with new visitors.

His strength was a secondary part of the image. What mattered was his solitude. We wanted to remove the most obvious external inputs from the scene and ask what remained: if that man developed cancer, how would we explain Venom's arrival in him?

The question forced us to look back. Perhaps arrival and action were separated by a long time. Perhaps the agent had accompanied him before his isolation, even from the beginning of life. His present solitude could coexist with an earlier biological history.

That was the island's contribution. It shifted attention from what a person acquires today to what they might have been carrying for much longer. It opened up prolonged persistence and transmission between generations as possibilities within the model.

Children and newborns had led to a related correction. Within the hypothesis, Venom had to be able to act early. Decades of deterioration could facilitate opportunities, but the idea also needed to encompass action at the beginning of life.

Ageing then retained another meaning: more time living with a possible agent, more changes in the terrain, and more occasions for favourable conditions to meet. The clock of arrival and the clock of action could run separately.

The twins returned, this time to help us think about their destinies. Two very similar people could share many of their conditions and experience different

events. To express that difference, a full stadium appeared, with a pigeon flying over it.

Eighty thousand people are seated. The pigeon soils just one. We can imagine the person affected as someone positioned along a particular trajectory, among many people exposed to the same scene. The image introduced chance and encounter into a conversation that risked turning every illness into a destiny written in advance.

Translated into the project, these images separate acquisition, persistence, and action. A person could harbour the agent long before it acted; another could have similar conditions and follow a different path. An inherited predisposition would describe the terrain; transmitting an agent between generations would describe its arrival. Each possibility occupies a different place.

The island asked where it came from. Age asked how much time it had. The pigeon asked why that encounter and not another. Together, the three images broadened the story we wanted to study.

WE KNOW

Children can also develop cancer. When comparing personal histories, it is useful to distinguish predisposition, exposure, and the time of disease onset.

WE PROPOSE

Venom could accompany the body from very early on and act at another time. Transmission between generations is a possibility that emerged when considering isolation and newborns.

IT PREDICTS

Early, prolonged persistence would direct us to look for presence before transformation. Transmission between generations would require studying a specific relationship across generations.

HOW WE WOULD TRY TO BREAK IT

Test arrival, persistence, and action separately, to learn which of those proposals finds support and which needs revision.

Chapter 7 Surveillance and the sneeze

The twins' mother had recognised a difference thanks to a sneeze. Returning to that scene brought up a natural question: could there be a similar opportunity for the body's defences?

Within the model, Venom's camouflage allows us to imagine a discreet stay in something the body recognises as its own. While that association remains stable, the difference would be difficult to notice. A transition could change the situation and offer a moment of exposure.

The conversation imagined the defences as surveillance taking advantage of that opportunity. NK cells were mentioned as possible participants. The image helped formulate a relationship between concealment, signal, and recognition.

In biological terms, the question is what difference the cell or agent would present during that transition and which part of the immune response could recognise it. Camouflage thus becomes a question about detectability; the sneeze, a possible time window.

Relationships between immunity and cancer have their own precedents. Shankaran and colleagues studied the role of lymphocytes and interferon gamma in tumour development and properties in mice [R7]. The NCI describes mechanisms of tumour evasion and treatments that act on the immune response [R8]. These provide a context from which to formulate more precise questions.

Our protagonist wanted to extend the image to spontaneous regressions. He imagined that a brief exposure of Venom might allow its recognition and that the body might also come to control the transformed cells. The conversation was trying to connect a fleeting window with a much larger consequence.

That sequence has two parts the project must study separately: the response to the agent and the response to the cells that have changed. The distinction

between initiator and consequence, which had helped us with the lighter, is useful here again.

Regressions documented in certain contexts of childhood neuroblastoma [R9] also remind us of the importance of identifying which disease and circumstances are being considered. The project can retain its question and seek the context in which it would make sense to examine it.

Immunosuppression offered the reverse side of the image: reduced surveillance might increase the opportunities for persistence or action that the model attributes to Venom. This possibility remained as a line of research linked to the body's defences.

For the first experiment, we took away a simpler idea: a brief transition deserves attention. The sneeze returns to its main role, which is to teach us when to look. Immunological research could be developed later, if a candidate emerges that allows it to be formulated precisely.

WE KNOW

Immunity participates in complex relationships with tumours [R7, R8]. Spontaneous regressions have been documented in specific contexts [R9].

WE PROPOSE

A transition in Venom could change its detectability and open an opportunity for immune recognition.

IT PREDICTS

This line of inquiry would look for a temporal relationship between the candidate's signal and a defined immune response.

HOW WE WOULD TRY TO BREAK IT

Separate the response to the candidate from control of tumour cells and examine whether known mechanisms explain the sequence observed.

Chapter 8 Who is the host?

If Venom needs a host, why would it kill the person harbouring it?

The question seemed contradictory until we changed scale. For the model, the immediate refuge would be Eddie, the cell. As long as cells were available, there would still be places to stay. The organism's death could be a consequence of the process that had been set in motion, even while cellular refuges continued to multiply within it.

That reading allowed us to return to the experiment with greater clarity. What we wanted to exhaust within a sample was precisely that availability: millions of cells, thousands, hundreds, ten, one. The question of the host brought us back to the count.

The conversation then broadened the idea: what if Eddie could also be another biological entity on which Venom acted? Until then, we had assigned every role to the cell: lodging, object of transformation, and bearer of its consequences. Perhaps those roles needed to be separated.

A known oncogenic virus could enter the scene as a viral Eddie: another entity susceptible to Venom's action. Instead of treating it solely as an explanation competing with the common root, the model imagined that it might receive a modification or information attributable to Venom. HPV was mentioned; its relationship with certain cancers is documented [R4].

The proposal extended Venom's possible field of action. It retained the intuition of a root virus and asked whether some consequences could arise indirectly, through another entity. The interest lay in the chain of transformation and what each link would retain.

This is where the signature began to take shape. Looking for the agent currently present was one thing; looking for an alteration that persisted afterwards was another. That second path would develop further when antivirals appeared as possible probes.

To keep the first study clear, the refuge count would continue to refer to viable cells in the sample. The extension to other entities was recorded as another line of inquiry, with its own questions about modification, persistence, and transmission of information.

The breadth of the project could be preserved without trying to fit all of it into a single test. As happens when developing a project, a general idea can open up several studies, each resolving a specific part. The initial core remained observing what happens when available cellular refuges disappear.

WE KNOW

HPV has a documented causal relationship with certain cancers [R4]. Lodging, transformation, and persistence of an effect are different relationships.

WE PROPOSE

Retaining the possibility that Venom acts on other biological entities as well as cells. The first test focuses on the viable cellular refuge.

IT PREDICTS

Extension to another agent would require an attributable, recognisable modification in it. The cellular test studies the transition linked to its own refuges.

HOW WE WOULD TRY TO BREAK IT

Evaluate each line of inquiry under its defined conditions and revise the attribution if the expected modification or refuge dependence fails.

Chapter 9 Turning a conversation into a project

An idea can be very clear to those who have spent time discussing it and seem incomplete to someone who receives only a few pages.

Our protagonist's biologist friend pointed out that gap. She found it difficult to see where the proposal was heading and how it intended to get there. The conversation contained connections that the dossier still left implicit.

That observation gave the book a task: reconstruct the journey. The personal question, the agent and its consequence, the refuge, the sneeze, and the loss of viability had to come together in a single explanation. The reader had to be able to follow them without having been present at every conversation.

Our protagonist brings an architect's perspective. He is organising an idea into a project: what he wants to investigate, which relationships he imagines, which conditions matter, and where he places the first observation. Technical development will need contributions from those who work with the samples and instruments.

The objections contribute to that construction. The specialist researcher's objection separated the interest in finding the initiator from an immediate therapeutic application. The biologist friend's called for a fuller explanation. The objection about the filter forced a review of a decision that restricted in advance what could appear.

There was also enthusiasm, shortcuts, and statements that were later corrected. The archive preserves that journey because it helps explain how we arrived here. The book organises it so that each image does its work and each question finds its proper place.

At first, there was much talk of proving it through reasoning. Then a more productive question began to emerge: what situation would allow us to observe a difference? This was an important step for the project. The

analogies had helped us imagine; now they could help us choose what to look at.

The isolated sample offered a setting. Deterioration provided a transition. The viable-cell count offered a variable to follow. Controls would allow what was observed to be compared with other explanations. These pieces were beginning to form a recognisable proposal.

The task of the following pages is to develop it. To retain the intuition of a common viral agent, bound the first study, and explain which results would justify continuing the search. There is an idea sustained for twenty-eight years and a new decision: to give it a form that can be examined outside this conversation.

WE KNOW

The dossier needs to make explicit the connections that remain implicit in a conversation.

WE PROPOSE

Organising the idea into a concrete search: observing a sample's transition as it loses viable cellular refuges.

IT PREDICTS

The experimental proposal must express an observable expectation and specify how it will be compared with other explanations.

HOW WE WOULD TRY TO BREAK IT

Ask specialists to review the design's ability to distinguish the alternatives and identify the conditions that could make it fail.

Chapter 10 A thousand possible hiding places

A thousand viable cells. A thousand possible hiding places. In the language of our analogy, a thousand possible Eddies. If one viable cell remains, one possible Eddie remains; at zero, none of those cellular refuges remains.

The phrase does not count viruses. It does not claim that every cell contains one or that all refuges are occupied. It counts possibilities under a rule of the model: the concealed state attributed to Venom depends on the availability of viable cellular lodging within the sample.

If a hundred remain, there are a hundred possibilities of that kind. If one remains, one remains. When no viable cells remain, that set of lodgings has been exhausted. The simplicity of the count is what brought order to a conversation that had opened too many paths.

The scientific translation begins with the conditional. If the agent is present; if it depends on cellular viability for the state we are seeking; if the loss of that support triggers a detectable transition; and if our instruments can observe it, the transition might reveal information. The count expresses only the second condition. The others are not included in the arithmetic.

This matters because a deduction can seem inevitable when we have omitted one of its steps. "It can no longer continue in the same way" does not imply "it will come out and we will see it." It could lose activity, degrade with the cell, or produce a change beyond the reach of observation. The test needs to recognise those alternatives before it begins.

At this point, the conversation identified a mistake in direction: keeping the tissue indefinitely in a comfortable state could preserve precisely the condition we wanted to see disappear. The purpose of this question is not to prolong the sample's life for its own sake. It is to observe its evolution as it loses viability, without turning a method of brutal destruction into the explanation for everything that appears.

This does not establish how to do it in a laboratory. How to study the transition, its compatibility with measurements, and its limits belong to specialist assessment. Nor does it imply that the loss follows a perfectly smooth curve. Different regions of a tissue may retain different states; some readings may fluctuate. The diagram 1000 → 1 → 0 is a conceptual structure, not data or measured kinetics.

Viable cells	Possible viable cellular refuges
1000	1000
1	1
0	None of that kind

Figure 1. Conceptual count of viable cellular refuges. It does not represent data or imply that every cell harbours the agent.

Choosing a thousand serves to make a reduction visible. It does not fix the number of patients, fragments, or samples in a study. When a thousand tumours were imagined in the reconstruction, the aim was diversity and repetition; an actual sample size would need to be justified by the minimum frequency to be detected and the quality of the observations.

The last cell therefore has a precise role. It is not a magical instant or the place where we force Venom to follow a script. It represents the last possibility of a defined type of refuge. The interest lies in everything that happens before, during, and after that possibility ceases to be detected.

The experiment does not yet know what it will find. It can say why it is looking at that process and which assumptions it is putting at risk. That difference turns the image of hiding places into a question a researcher can examine.

WE KNOW

A count of possible refuges is not a count of agents present. The absence of detected viability does not mean that all cellular matter has disappeared.

WE PROPOSE

Observing a sample during the reduction of viable cellular refuges. The central idea remains: 1000 viable cells = 1000 possible hiding places; 1 = 1; 0 = none of that kind.

IT PREDICTS

The experimental version adds a window of detectability in a predefined fraction of samples. Without that additional commitment, dependence on viability is not enough to predict a finding.

HOW WE WOULD TRY TO BREAK IT

Check whether deterioration destroys or renders unobservable what we intended to study, and whether the committed pattern fails to appear despite sufficient coverage. Do not turn every absence into a new way of hiding.

Chapter 11 The entire transition

At first, the final scene occupied too much space. One cell remained; the cell died; something had to come out.

The correction was to remove that obligation. We were assigning our timetable and our preferred form of manifestation to the unknown. If the idea was to observe a transition, it made no sense to attend only to the last second of the performance we had imagined.

The full sequence became: viable tissue, deterioration, last viable cells, operational zero, and the subsequent period. The sneeze could occur in any of those phases. There might also be no sneeze, or it might not be recognisable with the available techniques.

The scientific translation of "continuous observation" needs precision. It means seeking a sustained temporal record through modalities compatible with that monitoring. It does not mean obtaining every possible physical and molecular measurement from every cell at every instant. Some readings consume or alter material. A data sequence from related samples is not equivalent to observing the same intact cell throughout the process.

Viable tissue → Deterioration → Last viable cells → Operational zero → Subsequent period

Inside and outside. A temporal record with explicit limits.

Figure 2. The entire transition is part of the question. Zero is operational and does not require release at an exact instant.

Looking inside and outside was another central correction. A signal appearing around a cell may come from its rupture, a vesicle, or other processes. A signal that disappears inside and appears outside does not by itself demonstrate active departure by the agent. The temporal relationship helps raise questions, but the identity of what is observed must be established.

Nor is it enough to say that nothing moves. Viability cannot simply be reduced to visible movement. The responsible team would have to define convergent indicators, spatial coverage, and sensitivity. The result will be "we detect no viability under these criteria in this volume," not unlimited certainty about every corner of the sample.

That sentence may seem to diminish the force of zero. The opposite is true: it gives zero a meaning others can assess. An undefined zero allows too much to be claimed; an operational zero allows discussion of what was measured and what might have escaped detection.

Time also has more than one scale. Two samples may have spent the same number of hours outside the body and retain different levels of viability. A comparison by the clock can confuse effects of deterioration with biological differences. That is why both chronology and observed state matter. Neither should silently replace the other.

And the record includes absences. Interruptions, unobserved fields, doubtful signals, and periods without anomalies are part of what happened. If we preserve only the striking moments, the final montage will look more convincing than the actual study.

The experiment seeks a complete record of the transition. That record will allow us to recognise a signal, examine its meaning, and assess how much the observation has been able to answer.

WE KNOW

Every measurement has a temporal, spatial, and analytical scope. A change of compartment and a loss of viability do not by themselves identify a causal entity.

WE PROPOSE

Following the entire transition, attending to the cellular interior and the associated medium. Documenting operational zero, intervals without observation, and differences in quality.

IT PREDICTS

If there is a transition linked to refuge depletion, its relationship with viability state must be reproducible under defined rules. It need not coincide with the exact death of the last cell.

HOW WE WOULD TRY TO BREAK IT

Show that the supposed transition is explained by cell rupture, occurs equally in appropriate comparisons, or depends on gaps in the record. Distinguish a negative result from a window that was never observed.

Chapter 12 Chambers that share nothing

If we place many tumours together and find something common, we may have manufactured the coincidence ourselves.

The image of independent chambers arose to avoid that mistake. Each sample had to retain its own history, its own associated medium, and a record that was not mixed with the others. Separation was part of the question, not an organisational convenience.

Its scientific translation is independence of observations and traceability. A coincidence between samples is interesting only if we have investigated how they might share a cause unrelated to the phenomenon we seek. Drawing separate containers in a figure is not enough.

Two chambers may share reagents, equipment, processing, or contamination predating their separation. They may also share analytical decisions that produce the same artefact. Physical isolation reduces one route of confusion, but does not by itself certify that an entity has appeared independently.

The study by Zapatka and colleagues on viral associations illustrates the importance of this precaution. Their analyses included investigation of patterns compatible with contamination and effects linked to sequencing dates [R5]. Recurrence, even across projects, is not enough to establish tumour origin. That precedent matters especially to a project seeking something common.

Independence also has a biological dimension. Ten fragments of one tumour are not ten patients. Thousands of images are not thousands of independent origins. They may improve knowledge of a sample, but they do not multiply the number of people represented.

Tumour diversity remains useful: it forces us to ask whether the candidate's identity is preserved in different contexts. But diversity and size do not replace

quality. A large collection of samples with the same error can repeat that error with enormous precision.

The conversation then imagined a laboratory almost without people: closed chambers, remote observation, and automation. The idea expressed a wish to reduce interventions and preserve records, as well as to avoid assuming in advance that the unknown was harmless. Actual decisions on containment and safety would belong to the institution and the responsible specialists. The absence of a person beside the instrument does not by itself demonstrate that a system is safe.

Automation does not eliminate human interpretation either. Someone decides what is recorded, what triggers an alert, and which data are excluded. Those decisions can introduce expectations even when nobody touches a sample for hours.

That is why the relevant chamber is more than a container. It is a unit of observation with provenance, limits, and records. If a signal appears, we should be able to follow its history without gaps. If it does not appear, we should know what was actually observed.

Independence is documented by tracing the origin of every sample and every signal. This will allow us to assess the meaning of a coincidence across different tumours.

WE KNOW

A recurrent signal may come from a common source of contamination [R5]. Fragments and images from the same donor do not constitute independent donors.

WE PROPOSE

Separate samples, documented provenance, and comparisons distributed across batches and contexts. Automation is assessed as a monitoring tool, not an absolute guarantee.

IT PREDICTS

A candidate of biological interest must retain its pattern in new donors and independent conditions. It should not depend solely on one batch, day, or instrument.

HOW WE WOULD TRY TO BREAK IT

Actively seek a shared history that explains the coincidence. If the candidate follows the reagent or processing rather than the biological comparison, it loses its attribution to the proposed phenomenon.

Chapter 13 The experiment defends itself against us

A dying cell does not leave an empty, orderly scene. Its end produces material, changes, and structures that can attract attention.

The objection was immediate: if anything unusual is named Venom, the experiment will not discover an agent; it will discover our readiness to recognise it everywhere. That is why controls appeared.

The first comparison was healthy tissue of the same type. The second, a tumour produced experimentally through known mechanisms. The third, the naturally occurring tumour that prompted the question. The scientific translation does not consist of declaring three perfectly equivalent worlds, but of identifying which alternative each one helps examine.

Non-tumour tissue can help distinguish a phenomenon associated with general deterioration from one related to the tumour state. But the model allows lodging in apparently normal cells. It therefore cannot simultaneously decree that all healthy tissue is necessarily free of Venom. A signal in the control does not automatically kill the idea; an indistinguishable signal does not confirm a special relationship with the tumour either. The expected difference must be formulated beforehand.

The induced tumour allows us to ask whether a transformation with a known mechanism is sufficient to produce the same pattern. There are human organoid models in which defined alterations generate tumour properties; Drost and colleagues described an example in 2015 [R10]. This provides a concrete experimental alternative to requiring an unknown initiator. It does not simply become a universal certificate of the absence of every entity not sought.

This comparison helps specify the scope of the common root. The induced tumour represents a transformation obtained through a defined pathway; the naturally occurring tumour raises the question of its history of origin. A claim

encompassing all cancer, including experimental cancer, would have to explain both cases. The initial test must declare that scope so the comparison has a clear meaning.

A similar-looking signal can have ordinary explanations. Atkin-Smith and colleagues documented vesicle formation during apoptosis [R11]. Bullock and colleagues showed errors in identifying subcellular structures as viral particles by electron microscopy [R12]. These precedents do not tell us what we will find in Venom. They explain why visual resemblance needs verification.

These comparisons are supplemented by material and processing blanks. Their conceptual function is to ask whether the signal could come from the observation system. It also matters that tumour and control are compared in interpretable states: equal time does not guarantee equal deterioration or equal quantities of debris.

Imagine three possible results. A particle appears in all material as it degrades: perhaps it describes cell death. It appears only in one batch, including its blanks: perhaps it describes contamination. It appears in new tumours, is characterised concordantly, and retains a difference from appropriate controls: it deserves further investigation. None of these examples represents data obtained in the project.

The last scenario is more interesting, but still does not prove causality. A common component of tumours may be a consequence of their state. A control protects against specific errors; it does not turn an association into a cause merely by being called a control.

WE KNOW

Cell death can produce vesicles [R11], and images can lead to misidentification [R12]. Models of known transformation raise an objection that universality must address [R10].

WE PROPOSE

Comparisons with a defined purpose: non-tumour deterioration, transformation through a known mechanism, naturally occurring tumours, and technical sources of signal. Negativity of all healthy tissue is not assumed.

IT PREDICTS

The confirmatory version must commit to an interpretable differential pattern, as well as identity and recurrence. "Something appeared" is insufficient.

HOW WE WOULD TRY TO BREAK IT

Explain the candidate through cell death, contamination, a known tumour state, or comparison bias. A control that contradicts the expectation is not discarded simply because it is inconvenient.

Chapter 14 Not drawing the unknown in advance

A 20-200 nanometre filter appeared in one of the initial diagrams.

It seemed like a concrete, reasonable technical decision. It also revealed a contradiction: we were saying that we were looking for something unknown and then imposing a size on it before observing it. The tool was beginning to decide which answers could exist.

The correction was expressed in a sentence that must be preserved: "Use the hardware of current knowledge without imposing the software of our expectations on the unknown phenomenon."

Its scientific translation does not consist of abandoning filters, methods, or knowledge. Every measurement selects something. An instrument has limits of resolution, sensitivity, and compatibility. The rule asks us to recognise those selections and avoid turning an imagined characteristic of the agent, without justification, into the definition of what deserves attention.

That is why the range was withdrawn as a condition for the candidate. It was not replaced by the promise of a machine that sees every entity. A broad search still has blind spots. The honest way to extend its scope is to combine complementary questions and techniques, documenting what each can and cannot answer.

The history of viral detection offers real precedents. Epstein, Achong, and Barr described viral particles in Burkitt's lymphoma lymphoblasts in 1964 [R13]. That work shows that observing biological material can lead to important questions about an agent. It offers a precedent for viral observation that we can place alongside our question about a different time window.

Broad searches also exist. Zapatka and colleagues studied viral associations within PCAWG [R5]. It is legitimate to distinguish that design from the monitoring over time proposed here. These are different questions and

observation windows, which should be compared in terms of what each allows us to study.

Explants, for their part, are not an idea foreign to tumour research. Majumder and colleagues studied an ex vivo platform for assessing drug responses and heterogeneity [R14]. It shares an interest in tumour material outside the body, but answers a different question. That precedent allows us to place our proposal within work on tumour material outside the body and specify the question we want to add.

Our protagonist's proposal retains its origin: Venom would be a virus or a similar biological entity. Broadening that formulation allows observation without confining the candidate to an anticipated size or appearance. The viral idea continues to give direction to the search; characterisation will allow its properties to be specified.

Not all data will carry the same weight. An isolated shape, sequence, or composition may belong to different elements of a mixture. Linking them to a single candidate requires work. The MISEV2023 recommendations are a useful reference for recalling the care needed when characterising vesicles and their origin [R15], not a manual that certifies any unknown particle.

The experiment must remain open to the unexpected and closed to premature inference. Both fit in the same sentence: do not decide beforehand what it will look like, and do not decide afterwards that any appearance proves Venom.

WE KNOW

There are precedents for viral observation, pan-cancer studies, and ex vivo analyses [R5, R13, R14]. None of those cited establishes the universal agent proposed in the MVI.

WE PROPOSE

A search whose scope is described without imposing an imagined size, sequence, or morphology as the sole criterion. Specialists will select techniques for specific questions.

IT PREDICTS

If a common entity is detected, its identifying features must converge through compatible evidence. The impression that it "looks like a virus" does not satisfy that prediction.

HOW WE WOULD TRY TO BREAK IT

Check whether different measurements actually belong to different objects, whether the candidate is known, or whether the signal depends on an instrumental selection that manufactured the pattern.

Chapter 15 How Venom could lose

A hypothesis that can never lose does not need an experiment. It only needs new explanations whenever something does not fit.

The conversation recognised that danger when it proposed that broad, independent, technically adequate series without a common phenomenon should seriously weaken the model. This was a necessary commitment. It still had to become something more precise than "if we look a lot and nothing appears."

The scientific translation consists of fixing which version is being tested. The broad causal hypothesis attributes a common root. The presence hypothesis assumes that the candidate remains in some of the samples. The refuge hypothesis links a state to cellular viability. The detectability hypothesis expects some transition to be measurable. A negative result can affect that conjunction; it does not automatically identify which component failed.

That difficulty does not justify keeping all the components unchanged. If the conjunction justified the study and fails under agreed conditions, the experimental version must be abandoned or explicitly modified. The modification will be a new proposal, not a retrospective victory for the previous one.

Before validation, a minimum relevant frequency, a range of tumour types, an expected difference from controls, and observation limits would have to be specified. The book does not invent values to create an appearance of precision. The pilot should help estimate them; the confirmatory series should assess them once fixed.

Discovery and confirmation do different jobs here. In discovery, signals can be explored and identity rules formulated. In confirmation, the question is whether those rules work in new data. Using the same collection to invent a pattern and announce that it has confirmed it produces misleading certainty.

Observable anomaly → Identity and characterisation → Recurrence in new samples → Independent replication → Causal investigation

Figure 3. Different levels of evidence. Passing one level does not replace those that follow; the search may stop or change direction.

A positive search result would be a real candidate, characterised concordantly, recurring across different donors and types, with a pattern against comparators that is not explained by the alternatives examined. Reproduction in new samples and another laboratory would increase its interest. Whether it causes the process, accompanies it, or results from it would remain to be determined.

An ambiguous result is not a tentative positive. It may be an image without a linked molecular identity, a difference that disappears when batches are considered, or a signal that does not recur. The correct interpretation is to identify and resolve the remaining uncertainty, not to attach the agent's name to it.

A non-evaluable result is not a negative either. If the time window was lost, contamination occurred, or the defined volume was not covered, we do not know what we wanted to know. But accumulating non-evaluable studies does not support the hypothesis; it shows that the design needs review.

An informative negative meets the quality criteria and does not find the committed pattern. Repeated independently, it may justify abandoning that version. To say then that the agent had always left or always becomes undetectable would withdraw precisely the commitment that made the test possible.

Even a promising candidate can lose without the whole causal question being resolved. If its supposed common features separate into different entities, the attributed identity fails. If the pattern follows a contaminant, its interpretation fails. Not all defeats are of the same magnitude, but all must be recorded.

Methodological success does not depend on saving the name Venom. It depends on obtaining an answer with a clear meaning, including the meaning that the current formulation does not hold up.

WE KNOW

An absence is informative only in relation to defined observation conditions and expectations. Detecting an entity and attributing causality to it are different questions.

WE PROPOSE

Separating exploration, validation rules, and replication. Recording in advance which commitments make the MVI version evaluable and which result requires its revision.

IT PREDICTS

The evaluable version commits to bounded common identity, recurrence, and detectability. If those commitments are not met in adequate series, that version loses support.

HOW WE WOULD TRY TO BREAK IT

Accept informative negatives and refuted candidates. Do not subsequently lower expectations, regroup different entities, or broaden the meaning of refuge to prevent the model from failing.

Chapter 16 The signature and antivirals

The machine may have left while the message remains.

The Venom signature arose from that possibility. If the agent modifies a cell, perhaps some of what it leaves behind will allow its intervention to be reconstructed. The question broadens: as well as trying to find it, we could study what different tissues retain after it has passed through.

In biological terms, we seek a recognisable alteration, information, or a mechanism that can be related to the agent's action. The image of the machine is useful again: comparing messages to discover what operation they might share.

The extension to other biological entities reinforced this route. If an oncogenic virus received a modification attributable to Venom, it too might retain a signature. The project was beginning to distinguish the agent's presence from the persistence of its intervention.

That was when antivirals appeared as possible probes. The question was intriguing: if a compound used against a virus produces an unexpected effect in a tumour, what mechanism is it affecting? Within the hypothesis, it might be interfering with something related to Venom or with a consequence it had left behind.

There are precedents for experimental antitumour activity of antiviral drugs. Gills and colleagues studied nelfinavir in cellular systems and animal models, with effects related, among other changes, to endoplasmic reticulum stress [R16]. That work provides an example of how a compound can open questions about cellular mechanisms beyond its initial indication.

The conversation imagined a matrix: tumour A with compound X, tumour B with X, tumour A with Y, tumour B with Y. Then compare responses and ask what they share. The matrix organised an investigation of mechanisms and

possible signatures; its development belongs in a specialist study, separate from treatment decisions.

For a response to be a useful signature, it would have to be characterised and related to the candidate. Comparisons with known cellular effects, general toxicity, and other tumour alterations would come into play. This is the work that would turn a clue into an examinable relationship.

Our protagonist immediately saw the problem of scope. Every conversation opened another possibility, and the first project was beginning to grow in too many directions. We had to choose where to begin.

Two paths therefore remained. The physical search would try to observe Venom or a signal linked to it during refuge loss. The pharmacological search would study possible signatures and shared mechanisms. Both retained interest, but the first had to remain easy to recognise.

The decision was to begin with the sample and its transition. The signature remained as a second route, ready to be developed if observations or earlier work offered a concrete reason to do so.

WE KNOW

Experimental antitumour effects have been described for certain antivirals, with cellular mechanisms requiring characterisation [R16].

WE PROPOSE

Retaining a second line of inquiry: studying possible signatures of Venom's action through comparable alterations and responses.

IT PREDICTS

A signature attributable to the agent should be reproducibly related to its action and distinguishable from general cellular effects.

HOW WE WOULD TRY TO BREAK IT

Examine whether the response is explained by a known target, general toxicity, or a tumour alteration independent of the candidate.

Chapter 17 Finding it would open up further work

The project's ambition reaches beyond an image under a microscope. Finding the possible root agent would allow us to begin studying it: how it relates to its refuge, what changes it produces, and what remains afterwards.

Enigma and the encrypted message had given shape to that purpose. Having the initiator would allow us to try to observe a transformation from beginning to end and compare its steps. It would also allow us to ask whether the same intervention has different consequences depending on the tissue.

The journey is organised in stages. First, an observation that deserves attention. Then, discovering what has been observed. Next, checking whether it reappears with a concordant identity in other samples and whether another team can recognise it. That foundation would allow the study of its causal role to begin.

Each stage prepares the next. The tumour sample offers a window onto what happens during its loss of viability; investigating the origin of the transformation will require additional questions about what happened before. The project has room for both tasks and places them in temporal order.

Then a possibility appeared that widened the horizon: prevention. If the model eventually identified a common agent and understood its intervention, perhaps it would be possible to act before it left its consequences.

Smallpox provided a powerful historical reference. Its eradication, declared by the WHO in 1980, shows how far sustained intervention against a specific viral disease could go [R17]. The image helped us think about the value of knowing the agent and its cycle.

The relationship between HPV and certain cancers offers another context for understanding the preventive interest of a viral cause [R4]. In Project Venom, that horizon remains linked to what is discovered about the candidate, its scope, and the possibilities for intervention.

Preventing a transformation and treating a disease already under way are different objectives. The specialist researcher's objection returns here with renewed usefulness: it helps organise what we might learn about the origin and what work would be needed to act on the consequences.

In the enthusiasm of the conversation, there was even talk of absolute prevention. The current version expresses that aspiration as a future possibility conditional on results. It preserves what motivated it: the desire for understanding a beginning eventually to change what happens afterwards.

This is why we draw the whole staircase even though the first task focuses on its first step. The search has a direction. Observe, identify, understand, and, if it becomes possible, intervene. It all begins with obtaining an observation that opens the next step.

WE KNOW

There are experiences of preventing disease linked to viruses, with specific conditions and scope [R4, R17].

WE PROPOSE

Organising the programme into observation, identity, recurrence, replication, causal study, and possible applications.

IT PREDICTS

Causal attribution would require a relationship between the candidate and the transformation that goes beyond explanations of accompaniment or consequence.

HOW WE WOULD TRY TO BREAK IT

Revise the role attributed to the candidate if, when studied, it turns out to accompany or result from the tumour process.

Chapter 18 A project that can be put to the test

At this point, the proposal can be expressed clearly. Our protagonist believes that different cancers might share a root agent, a virus or a similar biological entity. He calls it Venom. He attributes to it a capacity for camouflage and a relationship with biological refuges that could change during a sample's deterioration.

Observing the transition arises from that combination. If we want to find it, it makes sense to ask what happens while the lodgings the model attributes to it diminish. The count 1000 → 1 → 0 gives that question a simple form.

Turning it into a study requires resolving concrete issues. Which samples offer an appropriate opportunity? How should their viability be measured? Which part of the interior and surroundings can be followed? How can a signal's identity be preserved? Which comparisons will distinguish a candidate from ordinary deterioration?

That is the conversation the project wants to open with specialists. The architect contributes the idea and its structure; those who know the samples, techniques, and analysis will be able to assess feasibility and turn the proposal into a defined design.

There are five main points for review: the scope of the common root, the opportunity for detection, the identity of what is observed, the meaning of refuge, and the effects of the study system itself. The experimental chapters explain why each matters and what testing it would require.

The results are also distinguished. A real, recurrent candidate would justify further steps. An ambiguous signal would call for characterisation. A failure of coverage would require the study to be reviewed. An absence under adequate conditions would have to carry weight in assessing the version put to the test.

Accepting that possibility is part of building the project. An idea can inspire a search and change when information arrives. Comparisons and objections help that transformation have a recognisable foundation.

The first study retains a specific scope. It investigates a transition linked to cellular refuges. Extensions concerning transmission, immunity, other entities, and pharmacological signatures remain organised so they can be addressed later with their own questions.

In this way, the initial ambition can be sustained while beginning with a bounded task. Seeking a common root is the direction. Observing a sample as it loses viability is the first proposed setting.

Twenty-eight years of a personal question have led to this decision: explain the idea clearly and propose where to look. That is the work our protagonist wanted to put on the table.

WE KNOW

Assessing the design requires knowledge of samples, measurements, analysis, and interpretation of results.

WE PROPOSE

A specialist feasibility review that keeps the hypothesis clear and bounds the first test.

IT PREDICTS

The version assessed must commit to an opportunity for detection, identity rules, and an expectation of recurrence.

HOW WE WOULD TRY TO BREAK IT

Compare results with those commitments and decide, using explicit criteria, whether to continue, reformulate, or abandon that version.

Chapter 19 Zero

Let us return to the sample.

As long as it retains viable cells, it retains the places to which the model attributes the possibility of harbouring Venom. The count falls and, with it, the situation we want to observe changes.

A thousand viable cells: a thousand possible hiding places.

A hundred. Ten.

One viable cell: one possible hiding place.

Zero viable cells: no viable cellular refuge within the sample.

We have watched throughout the journey. When the tissue was still alive, when it began to deteriorate, when few cells remained, and when the zero defined by our measurements was reached. The subsequent period is also part of the observation.

What matters is that continuity. The "sneeze" could occur before the final instant. It could manifest inside the cell or in its surroundings. That is why attention follows the transition instead of waiting only for a final scene.

The question that brought us here was simple: if Venom depends on those refuges to remain hidden, what happens when they cease to be available? The study is designed to record an answer, even if it forces us to change the idea with which we began.

The journey began long before the chambers. A father, seven years without smoking, two months between diagnosis and death. A question that accompanied our protagonist for twenty-eight years and has now found the form of a project.

The lighter separated the initiator from its consequence. Eddie gave camouflage a place and gave an identity to the cell that could be transformed without becoming Venom. The beaches introduced context. The isolated man

took our gaze back to the beginning of life. Enigma explained the interest in understanding transformation. The twins and the sneeze taught us to pay attention to a brief moment.

All those images converge here: maintain observation while the places where we imagine it can take refuge disappear.

WE KNOW

The study's zero is defined through criteria for viability, coverage, and sensitivity.

WE PROPOSE

Observing the entire transition, including when the last viable cellular refuge we can detect disappears.

IT PREDICTS

The experimental expectation is an opportunity for detection linked to that change in conditions, assessed under the agreed rules.

HOW WE WOULD TRY TO BREAK IT

Preserve the complete record and examine it even when it contradicts the initial expectation.

One viable cell remains.

Then it ceases to be viable.

That is the centre of the proposal. We have built a question and a place from which to try to answer it.

Let us be watching when the last one disappears.

Reference material

A glossary for returning to the text

Venom. Name of the proposed root agent: a virus or a similar biological entity, capable of initiating transformations and remaining in or leaving its refuge. It is the project's central hypothesis.

MVI. Itinerant Virus Model (Modelo del Virus Itinerante). Proposes a common agent capable of staying or leaving after a transformation. Itinerancy cannot be inferred from an absence and needs its own evidence.

Eddie. Name of the cell in the central analogy. It can harbour Venom and be transformed: venomised Eddie is still Eddie. The conversation later extended that role to other possible biological entities, including a virus.

Host cell. In the central experimental version, a cell that can function as a viable refuge. In extensions of the conversation, this was broadened to other entities; that extension is not silently introduced into the cell count.

Transformed cell. The cell that, within the narrative, would have undergone an alteration. It remains a cell. Its cellular identity is preserved during the proposed transformation.

Enigma. Analogy distinguishing the encoding machine from the encoded message. Venom takes the place of the machine; venomised Eddie, that of the message. Finding the possible initiator would allow us to try to study how it transforms Eddie, even if its effects persisted after its departure.

Beach. Analogy for the context that could affect establishment, action, or persistence. To become a scientific variable, it needs measurements; "calm" and "rough" are not already demonstrated clinical categories.

Sneeze. Transient event that could reveal a difference. It does not require physical departure or identify the cause of a signal.

Venom signature. Alteration the model imagines as a persistent consequence of its action. Attribution requires specificity and causality; a common tumour characteristic is not enough.

Hit-and-run. Initial intervention whose persistence is not necessary to maintain an effect. It does not automatically mean a complete agent travelling to another cell.

Viability. Cellular state assessed through defined criteria. Its loss does not imply physical disappearance of the cell or its components.

Operational zero. Absence of detected viability in the agreed volume and at the agreed sensitivity. It bounds an observation, not unlimited certainty about everything that might exist.

Candidate entity. Provisionally characterised object or component that deserves investigation. "Candidate" keeps its identity and relationship with the phenomenon open.

Recurrence. Reappearance under comparable identity rules. It must be distinguished from repetition of a common source of error.

Independent replication. Reproduction of the result with new samples and sufficiently independent conditions, ideally in another laboratory. It is not equivalent to repeatedly reanalysing the same file.

Causality. Relationship through which a factor contributes to producing an effect. Coexistence, resemblance, and pharmacological response are not enough on their own to attribute it.

Informative negative. A result without the committed pattern, obtained while meeting the conditions needed to look for it. It can weaken the version assessed.

Non-evaluable. A result that does not allow the question to be answered because of a specific problem with coverage, quality, or interpretation. It neither supports the hypothesis nor demonstrates biological absence.

What changed along the way

The archive is not rewritten to make the idea appear to have been clear from the start. There were extensions, objections, and corrections. The following are the most important when reading earlier versions without confusing them with this edition.

From the tumour as agent to the tumour as consequence. The first question imagined how a virus could be related to very different cancers. The lighter and the distinction between Venom and Eddie separated the postulated initiator from the transformed cell and its descendants.

From obligatory departure to observing a transition. The agent does not have to leave when the last cell dies. The conceptual study attends to the whole process, inside and outside, without deciding in advance what form a signal would take.

From the 20-200 nm filter to a declared scope. The range is withdrawn as a prior definition of the unknown. Observation without limits is not promised; the limits of the selected techniques must be explained.

From absolute camouflage to a commitment to detectability. An entity invisible in every imaginable situation cannot be tested through the proposed search. The evaluable version needs to assume some concrete opportunity for observation.

From finding to curing in a single leap, to separate stages. Finding something does not identify its cause, does not by itself allow a tumour to be reversed, and does not guarantee prevention. Historical statements of absolute cure or prevention are withdrawn as conclusions.

From supposedly clean controls to coherent expectations. Allowing lodging in apparently normal cells prevents us from simply requiring absence in all healthy tissue. The comparison must define which pattern would distinguish the hypothesis from alternatives.

From separate chambers to documented independence. Physical separation does not eliminate prior contamination, shared reagents, or common artefacts. The origin of a coincidence is investigated.

From zero cells to zero defined refuges. The count refers to viable cellular refuges. Extension to other biological supports is developed as a line of inquiry with its own question and its own test.

From historical analogies to relevant evidence. Fleming is retained as part of the conversation, not as proof of the MVI. The claim about Navier-Stokes remains in the archive as unverified and does not support this edition.

From a hardware diagram to measurable questions. The diagram in the historical letter is not adopted as a validated specification. Its equivalences between techniques and its claims of capability require specialist review.

Provenance of the materials

The main narrative source is the reconstruction "Viral hypothesis of cancer," blocks 0-32. These are seven complete assistant messages developed from the dialogue with our protagonist. Their texts and identifiers have been preserved in the master backup. Recovery of those blocks does not certify that the entire earlier conversation history has been exported.

The historical letter addressed to the specialist researcher and the epistemological debate were recovered from the VIRUS2 folder. Their categorical formulations and diagrams remain in their originals. The objections attributed to third parties in those materials are distinguished from independent access to their original communications.

The two MVI dossiers dated 3 October 2026 were incorporated as background. The more recent one already separated search, characterisation, validation, and causality, and made the limitations of operational zero explicit. The book continues that work without replacing its historical files.

The separate master document brings together the complete transcript, subsequent notes, letter, and historical debate. Its purpose is preservation. This book has a different purpose: to allow the idea, its evolution, and its experimental question to be followed in a continuous reading.

Correspondence with the original blocks

The table allows readers to return from the book to the reconstruction. Block numbering belongs to the archive; chapter numbering, to this edition. Titles are adapted here to identify content without participants' personal names; Eddie retains its name and conceptual function; the backup preserves the original titles.

Block	Archive content	Chapter
0	Before Venom	1
1	Separating the cause from the consequence	2
2	Eddie	3
3	The beaches	4
4	The fugitive	2
5	Enigma	5
6	The twins and the sneeze	3
7	Taking away its hiding places	10, 11
8	The experiment begins to defend itself against us	13
9	Not telling Venom what it must look like	14
10	Eddie might not be a cell	8
11	The signature	16
12	The immune system	7
13	Age	4, 6

Block	Archive content	Chapter
14	The problem of children	4, 6
15	The identical twins return	4, 6
16	Why does Venom kill its host?	8
17	Stop trying to prove it through reasoning	9, 15
18	Building the trap	12
19	Not waiting only for death	11
20	Looking inside and outside	11
21	The laboratory without people	12
22	What finding "something" would mean	13, 15, 17
23	The experiment can also kill it	15
24	The specialist researcher enters the story	5
25	The biologist friend challenges us from another angle	9
26	The virus that could also be Eddie	8
27	Antivirals as probes	16
28	What if finding it made prevention possible?	17
29	The mistake of keeping it too comfortable	10
30	What we are really looking for	14
31	The final question	10, 19
32	Zero	19

Annotated references

Sources consulted for this edition on 7 October 2026. The selection is targeted, not a systematic review of all the literature on cancer and viruses. The references support only the claims to which they are linked. DOIs and links allow the papers to be located; their inclusion does not imply their authors' endorsement of the MVI.

R1

National Cancer Institute. **What Is Cancer?**

https://www.cancer.gov/about-cancer/understanding/what-is-cancer

General context on cellular alterations, cancer development, and metastasis. It helps situate the additional hypothesis without presenting known mechanisms as mere descriptions with no explanatory value.

R2

National Cancer Institute. **Harms of Cigarette Smoking and Health Benefits of Quitting.**

https://www.cancer.gov/about-cancer/causes-prevention/risk/tobacco/cessation-fact-sheet

Supports the distinction between ending an exposure and immediately eliminating risk. It is not used to retrospectively diagnose the personal case in the first chapter.

R3

National Cancer Institute. **Risk Factors: Tobacco.**

https://www.cancer.gov/about-cancer/causes-prevention/risk/tobacco

Reference on the causal role of tobacco. The firewood-and-match metaphor does not replace that knowledge with an unproven viral requirement.

R4

National Cancer Institute. **HPV and Cancer.**

https://www.cancer.gov/about-cancer/causes-prevention/risk/infectious-agents/hpv-and-cancer

An established example of the relationship between viral infection and cancer, and of prevention through vaccination. It provides the scientific context for the HPV mentioned in the book.

R5

Zapatka M and colleagues. **The landscape of viral associations in human cancers.** Nature Genetics. 2020;52:320-330.

https://doi.org/10.1038/s41588-019-0558-9

Pan-cancer precedent within PCAWG: a collection of 2658 cancers of 38 types and a consensus cohort of 2656 donors for pathogen analysis. It is relevant for its viral associations and attention to contamination. It does not track viability to zero or demonstrate Venom.

R6

Shen Y, Zhu H, Shenk T. **Human cytomagalovirus IE1 and IE2 proteins are mutagenic and mediate "hit-and-run" oncogenic transformation in cooperation with the adenovirus E1A proteins.** Proceedings of the National Academy of Sciences. 1997;94:3341-3345.

https://doi.org/10.1073/pnas.94.7.3341

An experimental precedent in rat cells for transient viral intervention associated with transformation. Its scope cannot simply be extended to human tumours or a universal root agent.

R7

Shankaran V and colleagues. **IFNγ and lymphocytes prevent primary tumour development and shape tumour immunogenicity.** Nature. 2001;410:1107-1111.

https://doi.org/10.1038/35074122

Experimental evidence in mice on immunity and tumours. It does not identify a response to Venom or prove that NK cells explain the sequence imagined in the conversation.

R8

National Cancer Institute. **Immunotherapy for Cancer.**

https://www.cancer.gov/about-cancer/treatment/types/immunotherapy

Context on relationships between immunity, tumour evasion, and treatment. It is used to prevent the mother analogy from being confused with a complete description of the immune system.

R9

National Cancer Institute. **Neuroblastoma Treatment (PDQ), Health Professional Version.**

https://www.cancer.gov/types/neuroblastoma/hp/neuroblastoma-treatment-pdq

Documents spontaneous regression in specific contexts of childhood neuroblastoma. Its occurrence does not allow a single mechanism to be attributed to all tumour regressions.

R10

Drost J and colleagues. **Sequential cancer mutations in cultured human intestinal stem cells.** Nature. 2015;521:43-47.

https://doi.org/10.1038/nature14415

An experimental model of defined changes and tumour properties in human organoids. It helps discuss controls involving known transformation and the requirement for an unknown universal initiator.

R11

Atkin-Smith GK and colleagues. **A novel mechanism of generating extracellular vesicles during apoptosis via a beads-on-a-string membrane structure.** Nature Communications. 2015;6:7439.

https://doi.org/10.1038/ncomms8439

An experimental example of vesicle formation during cell death. It supports a concrete alternative to interpreting a particle as a new agent.

R12

Bullock HA, Goldsmith CS, Zaki SR, Martines RB, Miller SE. **Difficulties in Differentiating Coronaviruses from Subcellular Structures in Human Tissues by Electron Microscopy.** Emerging Infectious Diseases. 2021;27:1023-1031.

https://doi.org/10.3201/eid2704.204337

Analyses identification errors in microscopy. It provides a specific methodological warning about visual resemblance; it does not demonstrate that every candidate is an artefact.

R13

Epstein MA, Achong BG, Barr YM. **Virus particles in cultured lymphoblasts from Burkitt's lymphoma.** The Lancet. 1964;1(7335):702-703.

https://doi.org/10.1016/S0140-6736(64)91524-7

A historical precedent for viral observation in material related to a tumour. It is cited for its specific scope, without turning it into evidential precedent for a virus common to all cancers.

R14

Majumder B and colleagues. **Predicting clinical response to anticancer drugs using an ex vivo platform that captures tumour heterogeneity.** Nature Communications. 2015;6:6169.

https://doi.org/10.1038/ncomms7169

A precedent for functional research with ex vivo tumour tissue. Its question and results are not equivalent to the transition towards zero refuges proposed in the MVI.

R15

Welsh JA and colleagues. **Minimal information for studies of extracellular vesicles (MISEV2023): From basic to advanced approaches.** Journal of Extracellular Vesicles. 2024;13:e12404.

https://doi.org/10.1002/jev2.12404

Recommendations on vesicle characterisation and study. Used as guidance on the rigour required when attributing properties and provenance to particles, not as validation of an unknown candidate.

R16

Gills JJ and colleagues. **Nelfinavir, a lead HIV protease inhibitor, is a broad-spectrum, anticancer agent that induces endoplasmic reticulum stress, autophagy, and apoptosis in vitro and in vivo.** Clinical Cancer Research. 2007;13:5183-5194.

https://doi.org/10.1158/1078-0432.CCR-07-0161

A preclinical precedent for antitumour effects of an antiviral drug. It serves to raise questions about multiple mechanisms; it does not establish a Venom signature or a therapeutic recommendation.

R17

World Health Organization. **Smallpox.**

https://www.who.int/news-room/questions-and-answers/item/smallpox

Context on the eradication of smallpox and the role of vaccination. It illustrates a historical possibility conditional on a specific biology and intervention, not a general promise of eradication.